

Two-Factor RFID + Face Recognition Door

Master Engineering Package - Requirements, Specification & Design Baseline

Document ID	DOC-00	Status	Implementation Baseline
-------------	--------	--------	-------------------------

Purpose: One coherent reference for building the MVP without overengineering it. DOC-01 is normative for requirements; DOC-02 is the architectural model; DOC-03 is the implementation contract.

1. Executive Summary

This project demonstrates physical access control using two independent factors: possession of an enrolled RFID card and presentation of the corresponding enrolled face. The ESP32/MFRC522 handles device-side sensing and actuation. The laptop handles face recognition, the 10-second session window, authorization policy, persistence, and logging.

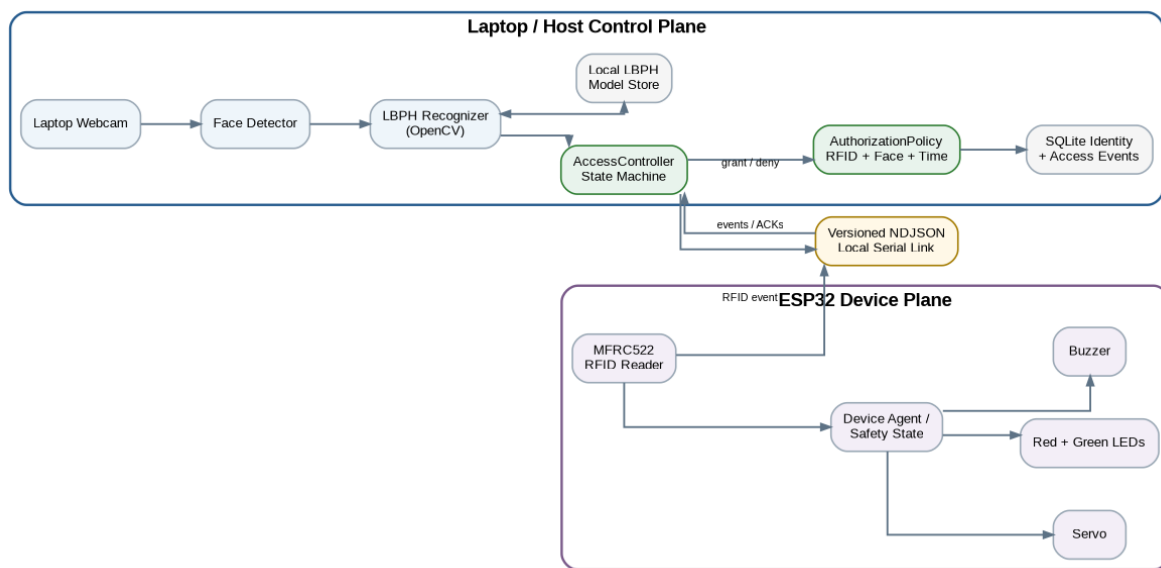


Figure 1. System architecture

2. System Contract

Property	Canonical value
Factors	RFID card + matching face
RFID component	ESP32 + MFRC522 + cards
Vision	Laptop webcam + OpenCV LBPH
Verification window	10 seconds after valid RFID

Property	Canonical value
Positive output	Servo opens + green LED + short beep
Negative output	Servo remains closed + red LED + long buzz
Primary user population	Small local set; typically 1-2 enrolled people for MVP demonstration
Persistence	SQLite + local face/model files
Host-device link	Versioned NDJSON over local USB serial
Security posture	Prototype; fail closed; no liveness; RFID UID not cryptographic credential

3. Canonical Decision

GRANT iff:

card is active and enrolled
AND now < session.deadline
AND exactly one usable face is visible
AND LBPH distance <= configured threshold
AND recognized identity == card owner
AND stable match count >= configured N

Otherwise:

DENY / TIMEOUT / FAULT
-> servo closed

4. Documents in This Package

Document	Use
DOC-01-SRS	Requirements, acceptance criteria, security limitations, traceability.
DOC-02-HLD	Components, boundaries, data ownership, failure handling, scalability seams.
DOC-03-LLD	Modules, interfaces, protocol, pin baseline, pseudocode, implementation constraints, test pyramid.

5. Build Sequence

1. Bring up ESP32 + MFRC522 and prove stable card UID reads.
2. Bring up servo, LEDs, and buzzer with a hardware-only device test.

3. Implement versioned serial protocol and a device simulator.
4. Implement SQLite identity store and RFID enrollment.
5. Implement webcam capture and face detection.
6. Implement LBPH enrollment and model persistence.
7. Implement AccessSession and AuthorizationPolicy as pure/testable code.
8. Integrate real hardware through adapters.
9. Calibrate the LBPH threshold using actual enrolled samples and the actual webcam.
10. Execute ACC-001..ACC-010 and record evidence in the test log.

6. Scalability Strategy

The design is scalable at the architecture boundary level but intentionally small at the deployment level. The core policy never depends on SQLite, a specific camera, a specific recognizer, or a specific device link. That permits later evolution to multiple doors, stronger card technologies, better vision models, centralized persistence, or networked device gateways without rewriting the access-policy core.

Today:

Host -> Serial -> One ESP32 -> One door

Later, if justified:

Host Policy Core

-> Device Gateway -> Door Registry -> N devices

or

Policy Core -> Repository implementation -> Central store

or

Policy Core -> FaceRecognizerPort -> stronger recognizer

7. Explicit Non-Optimization Rule

Do not optimize for people who are not here: The first implementation should be easy to read, easy to run, and easy to test on one laptop and one door. Capacity for future growth is provided by interfaces and stable contracts, not by unused infrastructure.

8. Security and Honesty

- The project proves two-factor identity binding; it does not prove production-grade biometric security.
- Printed-photo attacks may bypass LBPH because no liveness/presentation-attack detection is implemented.
- RFID UID-based identification is cloneable and is not equivalent to secure smart-card cryptography.
- The physical door/servo mechanism, enclosure, power, and manual bypass are outside the software assurance boundary.
- The safest software behavior on uncertainty is always closed door, not convenience.

9. Final Acceptance Gate

MVP passes: A live demonstration can show: correct card + correct face -> open; correct card + wrong face -> closed; unknown card -> closed; no face/slow face -> closed; communication or camera failure -> closed; and

the known printed-photo limitation is documented.

10. Reference Sources

OpenCV 4.13 LBPHFaceRecognizer API:

https://docs.opencv.org/4.13.0/df/d25/classcv_1_1face_1_1LBPHFaceRecognizer.html

NXP MFRC522 product page: <https://www.nxp.com/products/MFRC52202HN1>

NXP MFRC522 data sheet: <https://www.nxp.com/docs/en/data-sheet/MFRC522.pdf>

Espressif ESP32 UART documentation: <https://docs.espressif.com/projects/esp-idf/en/release-v5.2/esp32/api-reference/peripherals/uart.html>

Two-Factor RFID + Face Recognition Door

System Requirements & Software Requirements Specification

Document ID	DOC-01	Status	Implementation Baseline
-------------	--------	--------	-------------------------

Engineering baseline: Build the smallest complete system that proves two-factor identity binding. Do not add cloud, multi-door, distributed, or enterprise infrastructure unless a requirement below changes.

1. Document Control

Field	Value
Document ID	DOC-01-SRS
Version	1.0
Status	Implementation Baseline
Date	September 29, 2026
Primary audience	Implementers, reviewer, AI coding agent, test operator
System type	Embedded + local computer-vision access-control prototype

2. System Objective

The system shall open one physical door only when a valid enrolled RFID card and the corresponding owner face are presented in the required sequence. A valid card starts a 10-second verification window. A matching face causes a grant; wrong face, no face, timeout, ambiguity, or required-component failure causes denial or fault-safe closure.

Canonical authorization rule

```
AuthorizationDecision =
  active_rfid
  AND verification_window_valid
  AND exactly_one_usable_face
  AND lbph_distance <= threshold
  AND recognized_person == card_owner
  AND stable_match_count >= required_consistent_matches
```

3. Scope and Non-Goals

In Scope	Explicitly Out of Scope
One door; one ESP32; one MFRC522; a small set of enrolled people; laptop webcam; local host	Multiple doors; high user counts; multi-site deployment
RFID detection + local identity lookup	Cryptographically strong smart-card authentication
OpenCV LBPH face recognition + enrollment	Liveness detection; anti-spoofing; depth/IR sensing
10-second verification window	Cloud recognition; remote authentication; mobile app
Servo/LED/buzzer control	Enterprise IAM, Kafka, Kubernetes, service mesh, HA clusters
SQLite/local file persistence and local logs	Distributed database or internet-dependent operation

Why this boundary matters: The project is small. Scalability is achieved through replaceable interfaces and explicit contracts, not by prematurely deploying infrastructure the MVP does not need.

4. Actors and Operational Roles

Actor	Purpose
Primary authorized user	Owns an enrolled RFID card and enrolled face.
Friend / test user	Provides a distinct enrolled face for negative-path testing.
Operator / implementer	Enrolls identities, configures the system, runs tests, and inspects logs.
ESP32 device	Reads RFID and drives physical indicators/servo.
Laptop host	Runs recognition, policy, persistence, and device communication.

5. Functional Requirements

ID	Requirement	Specification
FR-001	Startup and safe state	System initializes all required components and remains closed until the controller is healthy.
FR-002	RFID detection	ESP32 detects RFID card presentation using MFRC522 and reports the card

ID	Requirement	Specification
		UID to the host.
FR-003	RFID validation	Host accepts only active enrolled RFID UIDs and associates them with exactly one person.
FR-004	Verification window	A valid RFID event starts a 10-second face-verification window using a monotonic clock.
FR-005	Camera capture	Host acquires frames from the laptop webcam during the verification window.
FR-006	Face detection	Host determines whether zero, one, or multiple faces are visible in a frame.
FR-007	LBPH recognition	Host uses OpenCV LBPHFaceRecognizer for recognition of the detected face.
FR-008	Recognition threshold	LBPH distance is evaluated against a configured threshold; rejected results never grant access.
FR-009	Identity binding	Recognized face identity must equal the person bound to the presented RFID card.
FR-010	Stable-match gate	A grant requires N consecutive acceptable matches for the expected identity; default N=3.
FR-011	Wrong-face denial	A detected enrolled face that does not match the RFID owner causes immediate denial.
FR-012	No-face/timeout denial	No usable matching face before 10 seconds causes timeout and denial.
FR-013	Grant actuation	Only ACCESS_GRANTED causes the ESP32 to move the servo to the configured open position.
FR-014	Door hold and close	After grant, the door remains open for the configured hold interval, then returns to the closed position.
FR-015	Denied indication	Denial illuminates the red LED and produces a long buzz; servo remains closed.

ID	Requirement	Specification
FR-016	Granted indication	Grant illuminates the green LED and produces a short beep.
FR-017	Versioned protocol	Host/device communication uses versioned newline-delimited JSON messages with correlation IDs.
FR-018	Command acknowledgement	ESP32 acknowledges each valid actuator command so the host can distinguish accepted and failed execution.
FR-019	Fail closed	Unknown, stale, malformed, duplicate, or unactionable access commands do not open the door.
FR-020	Enrollment	Authorized operator can enroll a person by capturing face samples and associating one or more RFID cards.
FR-021	Access logging	Each authorization attempt records timestamp, card UID, expected identity, observed outcome, and reason.
FR-022	Model persistence	LBPH model metadata and trained model are saved locally and reloadable after restart.
FR-023	Health state	Host and ESP32 expose enough health/status information to prevent grants while required components are unavailable.
FR-024	Safe recovery	After restart or communication loss, the physical output returns or remains in the closed-safe state.
FR-025	Known spoofing limitation	Documentation shall explicitly state that printed-photo and presentation attacks are not reliably detected by this MVP.

6. Non-Functional Requirements

ID	Attribute	Specification
NFR-001	Responsiveness	The system starts face verification immediately after valid RFID acceptance and provides prompt local

ID	Attribute	Specification
		feedback.
NFR-002	Timing	Verification window is 10 seconds; timing uses monotonic time to avoid wall-clock changes.
NFR-003	Reliability	The system shall fail closed on missing camera, lost device link, malformed commands, or recognition uncertainty.
NFR-004	Maintainability	Recognition, authorization, persistence, communication, and hardware control are isolated behind interfaces.
NFR-005	Determinism	The authorization policy is explicit and testable; a face-recognition result never directly controls hardware.
NFR-006	Observability	Important state transitions, outcomes, failures, and correlation IDs are locally loggable.
NFR-007	Scalability seam	Repositories, transport, camera, recognizer, and device identity are replaceable without rewriting the core policy.
NFR-008	Local privacy	MVP operation requires no cloud service; biometric samples remain on the local host by default.

7. Security Requirements

ID	Requirement	Specification
SEC-001	Two-factor correlation	Grant requires both valid RFID possession and matching face identity for the same person.
SEC-002	Closed default	Power-up, restart, communication loss, and controller fault must not result in door opening.
SEC-003	Command safety	Only an authenticated-in-process authorization decision may produce ACCESS_GRANTED for the device adapter.
SEC-004	Replay/stale handling	Access commands carry

ID	Requirement	Specification
		session/command identifiers and expire with the access session.
SEC-005	Single decision point	One AccessController/AuthorizationPolicy owns the final decision; adapters cannot grant access independently.
SEC-006	RFID limitation	MVP treats card UID as an identifier, not as a cryptographically strong credential; cloning is out of scope.
SEC-007	Biometric limitation	LBPH is not liveness detection; printed photos and other presentation attacks may succeed.

Security posture: This is a prototype, not a production security product. Common RFID UIDs can be cloned, the USB/serial channel is trusted locally, and LBPH is not a liveness detector.

8. State Model

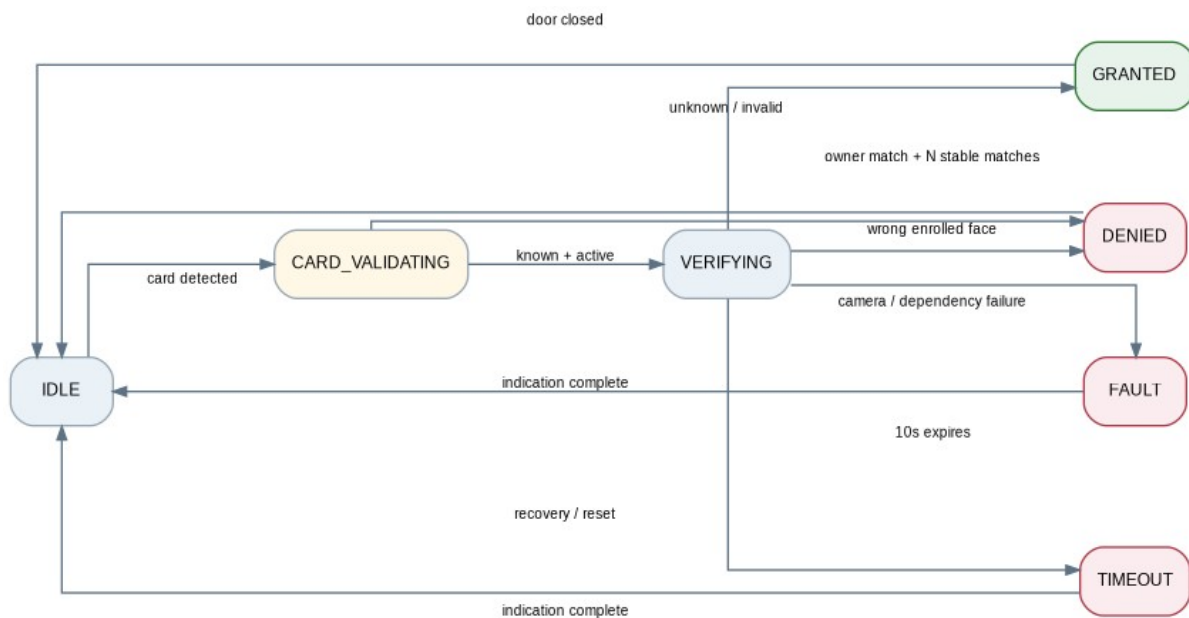


Figure 1. Canonical access-control state machine

State	Entry condition	Allowed outputs	Exit condition
IDLE	No active session	Servo closed; indicators off	RFID detected or fault
CARD_VALIDATING	RFID event received	No door open	Known active card -> VERIFYING; otherwise

State	Entry condition	Allowed outputs	Exit condition
			DENIED
VERIFYING	Known card + timer started	Camera capture only; no actuator grant yet	Owner match -> GRANTED; wrong face -> DENIED; 10s -> TIMEOUT; dependency failure -> FAULT
GRANTED	Authorization rule true	Open servo + green + short beep	Hold timer expires -> close -> IDLE
DENIED	Any explicit denial	Servo closed + red + long buzz	Indication complete -> IDLE
TIMEOUT	10s window expires	Servo closed + red + long buzz	Indication complete -> IDLE
FAULT	Required dependency unavailable	Servo closed; safe indication/log	Operator/system recovery -> IDLE

9. Access Flows

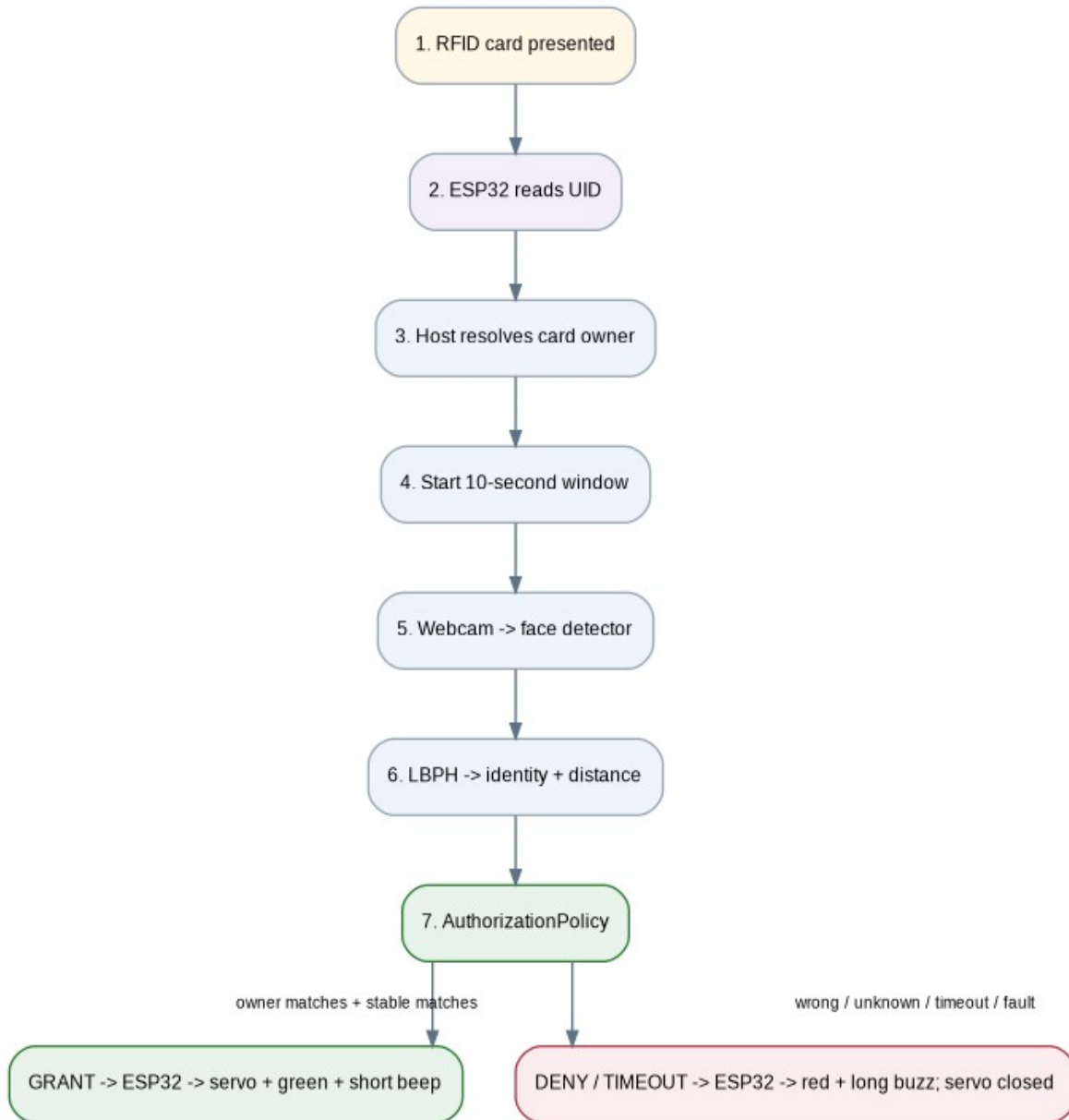


Figure 2. End-to-end authorization flow

10. Face Recognition Specification

The MVP uses OpenCV LBPHFaceRecognizer. The recognition stage is distinct from face detection: detection answers whether a face exists in a frame; LBPH answers which enrolled label is the closest match and exposes a distance/threshold mechanism. Recognition input is normalized grayscale face imagery.

```

frame -> detect face -> select usable face -> grayscale/normalize -> LBPH.predict -> (label, distance)
if distance > threshold: reject recognition result
if label != expected_face_label: deny
if label == expected_face_label for N consecutive samples: grant
  
```

The threshold is an empirical configuration value calibrated using the actual webcam, lighting, distance, and enrolled samples. The project shall not claim that one threshold is universally secure.

11. Enrollment Specification

1. Operator chooses a person identity.
2. Operator associates the RFID UID with that person.
3. Webcam captures multiple samples while a single face is visible.
4. Each sample is detected, cropped, converted to grayscale, and stored under the person identity.
5. LBPH model is trained/updated from the approved sample set.
6. Model metadata and identity-label mapping are persisted locally.
7. Operator performs a live verification test before considering the person active.

Enrollment rule: The system shall not treat a single captured photograph as a sufficient training set. Capture variation deliberately but keep the workflow simple for this MVP.

12. Protocol Requirements

Laptop and ESP32 communication shall use a versioned NDJSON protocol. Each message carries a type and, where relevant, event_id, session_id, and command_id so retries and stale commands can be identified.

ESP32 -> Host

```
{"v":1,"type":"rfid_detected","event_id":"evt-123","uid":"A1B2C3D4"}
```

Host -> ESP32

```
{"v":1,"type":"access_command","command_id":"cmd-456","session_id":"sess-123","decision":"grant","door_hold_ms":3000}
```

ESP32 -> Host

```
{"v":1,"type":"command_result","command_id":"cmd-456","result":"executed"}
```

ESP32 -> Host

```
{"v":1,"type":"heartbeat","device_id":"door-01","state":"CLOSED"}
```

13. Persistence Requirements

Entity	Required fields	Notes
Person	person_id, display_name, face_label, status, created_at	Identity used by both RFID and face bindings.
RFIDCard	rfid_uid, person_id, status, enrolled_at	MVP treats UID as identifier, not cryptographic proof.
FaceSample	sample_id, person_id, file_path, captured_at, quality/status	Local raw images used for training; operator controls enrollment.
SystemConfig	verification_window_s, door_hold_s, threshold, required_consistent_matches, camera_index, serial settings	Version/config validated at startup.
AccessEvent	event_id, timestamp, session_id,	Audit/debug record.

Entity	Required fields	Notes
	card_uid, expected_person, observed_person, result, reason	

14. Timing and Safety Rules

Rule	Specification
Verification window	10 seconds measured from valid RFID acceptance using monotonic host time.
Grant authority	Only AccessController/AuthorizationPolicy can produce a grant command.
Default physical state	Servo closed; no green success indication.
Failure state	Any unknown, stale, malformed, duplicated, or unacknowledged access command must not open the door.
Door hold	Default 3 seconds; configurable.
Stable matches	Default 3 consecutive acceptable LBPH matches.

15. Acceptance Criteria

ID	Scenario	Input	Expected Result
ACC-001	Correct identity	Enrolled card for Person A + Person A face within 10s	Green LED + short beep + servo opens
ACC-002	Wrong face	Person A card + Person B face	Red LED + long buzz + servo remains closed
ACC-003	Unknown card	Unknown card + any face	Denied without opening
ACC-004	No face	Person A card + no face for 10s	Timeout, red LED, long buzz, closed
ACC-005	Too slow	Person A card + correct face after 10s	Denied
ACC-006	Recognized wrong card	Person B card + Person A face	Denied
ACC-007	Multiple faces	Person A card + two visible faces	Denied; no grant
ACC-008	Camera failure	Valid card then camera	Denied/fault; door closed

ID	Scenario	Input	Expected Result
		unavailable	
ACC-009	Link failure	Valid card or grant while ESP32 link fails	No unsafe open; event logged
ACC-010	Photo spoof test	Person A card + printed Person A photo	May be accepted; limitation documented

16. Known Limitations and Security Disclosure

- A printed photograph may fool LBPH in this prototype. No liveness detection or presentation-attack detection is implemented.
- RFID card UIDs are identifiers, not strong cryptographic authentication. Cloning/replay resistance is out of scope for this MVP.
- The laptop is a trusted local control host. This is not a hardened physical security controller.
- Environmental changes in lighting, pose, camera angle, and enrollment quality can affect LBPH recognition.
- A servo mechanically attached to a door is only as secure as the physical enclosure and power design.

17. Traceability

Requirement group	Design artifact	Primary tests
FR-002..FR-004	Device Agent + AccessController + monotonic session timer	ACC-001..ACC-006
FR-005..FR-012	CameraPort + LBPH adapter + AuthorizationPolicy	ACC-001..ACC-007, ACC-010
FR-013..FR-016	Device command adapter + device firmware state	ACC-001..ACC-006, ACC-009
FR-017..FR-019	Protocol contract + command idempotency	ACC-008..ACC-009
FR-020..FR-022	EnrollmentService + repositories + model store	Enrollment integration tests
FR-023..FR-025	Health service + safety policy + documentation	Fault injection and photo test

18. References

OpenCV LBPHFaceRecognizer API: docs.opencv.org

NXP MFRC522 product page / lifecycle: [nxp.com](https://www.nxp.com)

NXP MFRC522 data sheet: [MFRC522 datasheet](#)

Espressif ESP32 UART documentation: docs.espressif.com

Two-Factor RFID + Face Recognition Door

High-Level System Design & Architecture

Document ID

DOC-02

Status

Implementation Baseline

Design objective: Keep one authoritative local control loop, isolate hardware and ML behind ports, and preserve future expansion seams without introducing distributed infrastructure into the MVP.

1. Architecture Overview

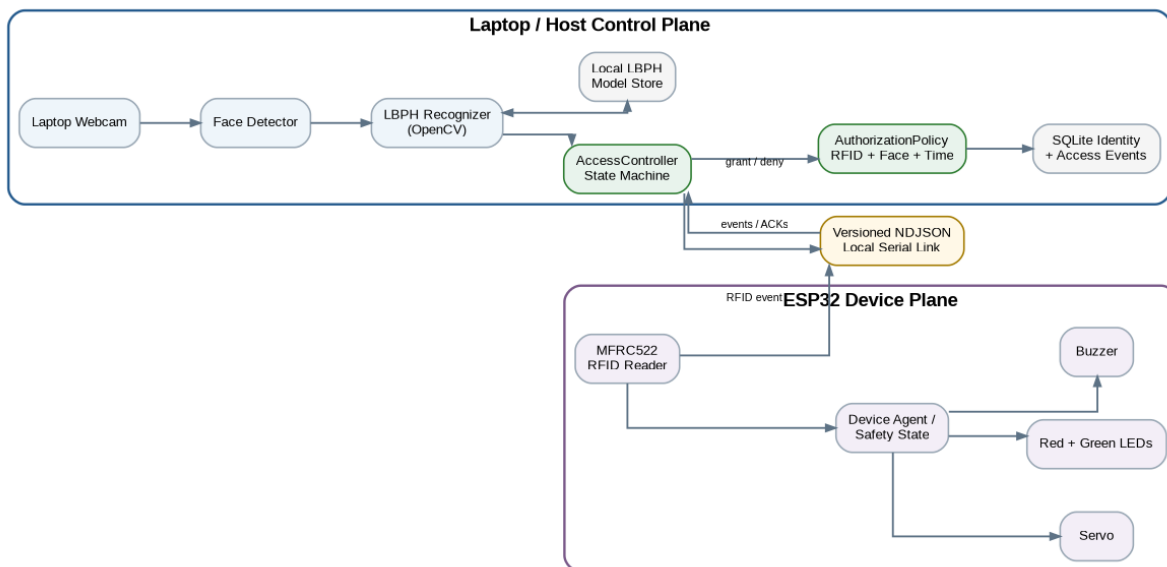


Figure 1. High-level component architecture

The laptop is the system brain. The ESP32 is a deterministic device agent. The core application makes the access decision; adapters execute the decision on hardware. This avoids coupling recognition code directly to servo control.

2. Architectural Principles

Principle	Decision	Reason
Single decision authority	AccessController + AuthorizationPolicy	Prevents an RFID handler, ML module, or device adapter from independently opening the door.
Local first	SQLite + local files + USB serial	Matches the real MVP: one door, one host, few users, no internet requirement.
Ports and adapters	CameraPort, FaceRecognizerPort,	Allows later technology changes

Principle	Decision	Reason
	DeviceTransportPort, repositories	without rewriting the domain policy.
Fail closed	Safe states at host and device	Physical access control must not open because of uncertainty or failure.
Explicit state machine	AccessSession state is persisted/logged in events	Makes timing, failure, and test behavior deterministic.
Scalability by seam	Device ID, repository interface, transport abstraction	Future scale becomes an extension of an existing boundary, not a rewrite of business logic.

3. Logical Layers

Layer	Responsibilities	Must not do
Domain	AccessSession, Identity, AccessDecision, ReasonCode, AuthorizationPolicy	Call OpenCV, serial, camera, filesystem, GPIO.
Application	AccessController, EnrollmentService, HealthService, EventLogger	Own low-level hardware details.
Ports	Interfaces for RFID, camera, recognizer, transport, clock, persistence	Contain implementation-specific behavior.
Adapters	OpenCV, SQLite, serial, filesystem, clock, model store	Make business authorization decisions.
ESP32 firmware	RFID I/O, actuator control, command validation, device-safe state	Recognize faces or decide identity authorization.

4. Runtime Interaction

1. ESP32 detects a card and emits rfid_detected with a unique event_id.
2. DeviceTransportAdapter delivers the event to AccessController.
3. AccessController resolves card ownership through IdentityRepository.
4. A valid card creates AccessSession(session_id, expected_person, deadline).
5. CameraPort provides frames while the session is VERIFYING.
6. FaceRecognizerPort produces RecognitionResult(label, distance) for detected faces.
7. AuthorizationPolicy evaluates identity binding, threshold, match stability, and time.
8. Only a GRANT decision is translated into an access_command to the device.
9. ESP32 validates and executes the command, then returns command_result.
10. EventLogger records the outcome and correlation identifiers.

5. Failure Handling Matrix

Failure	Host action	Device action	Result
Unknown RFID	Do not start face verification; log UNKNOWN_RFID	Remain closed	DENIED
No face	Keep scanning until deadline	No output change	TIMEOUT if deadline expires
Wrong enrolled face	Immediate deny	Red + long buzz; closed	DENIED
Multiple faces	Treat as ambiguous and deny	Red + long buzz; closed	DENIED
Low-confidence/threshold fail	Continue within 10s unless explicit wrong identity is known	No grant	Eventually TIMEOUT/DENIED
Camera failure	Enter FAULT; stop grant path	Remain closed	FAULT
Serial loss	Cancel/deny outstanding authorization	Remain closed	FAULT/DENIED
Malformed command	Log and reject	Ignore safely	No grant
Duplicate grant command	Idempotent by command_id	Do not re-trigger open unnecessarily	Safe
Restart	Reinitialize into CLOSED/IDLE	Servo closed target, indicators safe	Safe recovery

6. Authorization State Model

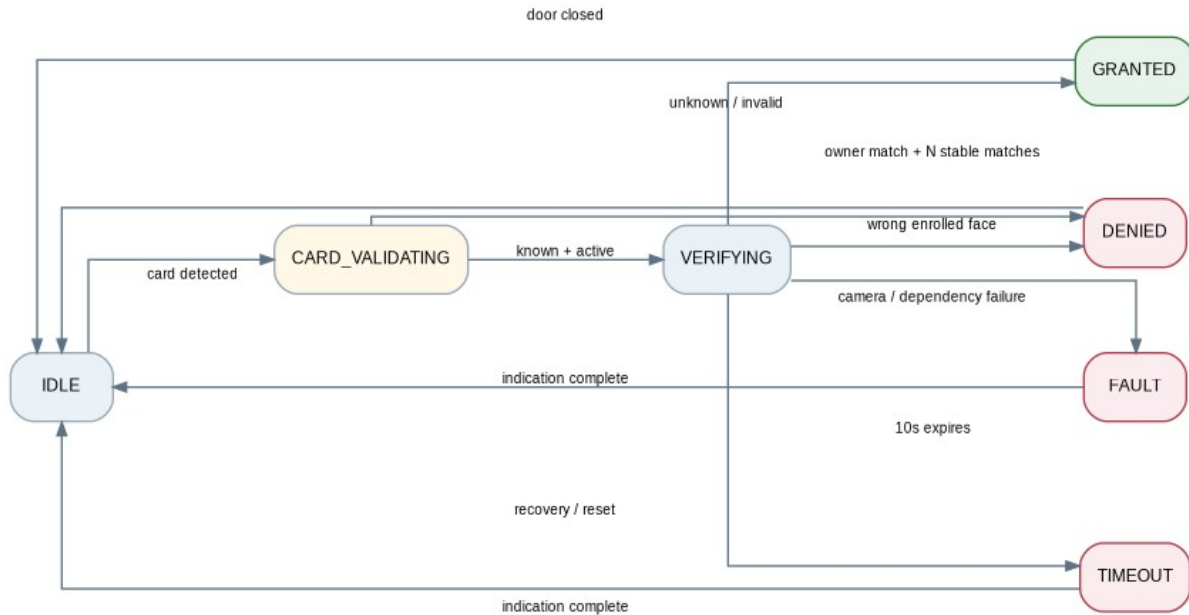


Figure 2. State ownership is centered in the host controller

7. Communication Boundary

For the MVP, use the USB serial path exposed by the ESP32 development board. The host sees a logical DeviceTransportPort, so the implementation can later switch to a network transport without changing the domain or authorization modules. Espressif documents configurable UART parameters and application use of UART interfaces.

```

TransportPort
  send(command: DeviceCommand) -> None
  receive() -> DeviceEvent | None
  health() -> TransportHealth
  
```

```

Message envelope
  v: integer
  type: string
  event_id/session_id/command_id: correlation identifiers
  payload: object
  
```

8. Scalable Boundary Map

MVP implementation	Future replacement without domain rewrite
SQLiteIdentityRepository	PostgreSQLIdentityRepository / remote identity service
Local JSONL/FileEventRepository	Append-only event store / centralized audit pipeline
OpenCVCameraAdapter	Network camera / video capture service
OpenCVLBPHRecognizer	Another local model or isolated inference service

MVP implementation	Future replacement without domain rewrite
Esp32SerialAdapter	Authenticated network device gateway
Single DeviceRegistry entry	Multiple-door device registry

Scale rule: Do not introduce a microservice until a real boundary requires independent deployment, ownership, scaling, or trust. The ports above are the seams that make such evolution possible later.

9. Data Ownership

Data	Owner	Storage
Person and RFID bindings	IdentityRepository	SQLite
Face training samples	EnrollmentService	Local filesystem
LBPH model + label map	FaceRecognizer adapter / ModelStore	Local filesystem
Access events	EventLogger / EventRepository	SQLite + optional JSONL export
Live session state	AccessController	In-memory; events persisted for audit
Physical device state	ESP32 Device Agent	ESP32 RAM; reported by heartbeat/status

10. Observability

- Every RFID event gets event_id.
- Every authorization attempt gets session_id.
- Every device action gets command_id.
- Logs include reason codes instead of free-form only messages.
- Heartbeat exposes device_id and physical safe state.
- Recognition diagnostics may log distance/threshold for development, but raw face images should not be copied into routine access logs.

11. Deployment Topology

MVP deployment

Laptop

- ├─ access-control host process
- ├─ local SQLite database
- ├─ local face samples / LBPH model
- └─ USB serial connection



12. Hardware Design Notes

A practical baseline pin map is given in the LLD. The MFRC522 is a 13.56 MHz contactless reader/writer IC and NXP currently marks the MFRC522 product family as End of Life / not recommended for new designs. It is therefore acceptable here as a fixed educational prototype component, but it should not be used as the basis for a new commercial hardware platform without lifecycle review.

Power integrity: Do not power a servo directly from a fragile logic rail. Use an appropriate 5V supply for the servo, tie grounds together, and keep the MFRC522 logic at its supported voltage. Poor servo power can reset the ESP32 and create misleading software failures.

13. Technology Choices

Area	MVP choice	Why
MCU	ESP32	Low-cost control of RFID and simple actuator I/O.
RFID	MFRC522	Fixed user requirement; simple reader integration for prototype.
Host language	Python	Fast local iteration and direct OpenCV integration.
Computer vision	OpenCV + LBPH	Explicit project requirement; deterministic, local, understandable.
Persistence	SQLite	Enough for a single local host and small user population.
Protocol	NDJSON over USB serial	Simple, debuggable, versionable; no broker/server required.

Two-Factor RFID + Face Recognition Door

Low-Level Design, Interfaces & Implementation Contract

Document ID	DOC-03	Status	Implementation Baseline
-------------	--------	--------	-------------------------

Implementation rule: This document is the direct handoff to the implementer or AI coding agent. Requirements are normative; design choices are the default implementation; deviations must preserve requirement IDs and acceptance behavior.

1. Repository / Module Layout

```

two-factor-door/
├── app/
│   ├── domain/
│   │   ├── models.py
│   │   ├── states.py
│   │   ├── decisions.py
│   │   └── reason_codes.py
│   ├── application/
│   │   ├── access_controller.py
│   │   ├── enrollment_service.py
│   │   ├── health_service.py
│   │   └── event_logger.py
│   ├── ports/
│   │   ├── rfid.py
│   │   ├── camera.py
│   │   ├── recognizer.py
│   │   ├── device_transport.py
│   │   ├── repositories.py
│   │   ├── model_store.py
│   │   └── clock.py
│   ├── adapters/
│   │   ├── esp32_serial.py
│   │   ├── opencv_camera.py
│   │   ├── opencv_lbph.py
│   │   ├── sqlite_repositories.py
│   │   ├── file_model_store.py
│   │   └── monotonic_clock.py
│   ├── config.py
│   └── main.py
├── firmware/
│   ├── src/main.cpp
│   └── protocol.h
├── data/
│   ├── app.db
│   ├── faces/<person_id>/sample_*.png
│   └── models/lbph.yml

```

```

| └─ logs/access.jsonl
└─ protocol/
|   └─ schema.json
└─ tests/
|   └─ unit/
|   └─ integration/
|   └─ hardware/
└─ docs/

```

2. Domain Model

Object	Responsibilities	Key fields
Person	Represents an enrolled identity	person_id, display_name, face_label, status
RFIDCard	Binds a physical card UID to a person	uid, person_id, status
RecognitionResult	Carries model output without granting access	label, distance, captured_at
AccessSession	Tracks one RFID-driven verification attempt	session_id, card_uid, expected_person_id, started_at, deadline, state, match_count
AccessDecision	Final authorization result	decision, reason, person_id, session_id
SystemConfig	Validates required runtime settings	window, hold, threshold, match count, ports

3. Interfaces / Ports

```

class CameraPort(Protocol):
    def open(self) -> None: ...
    def read(self) -> Frame | None: ...
    def close(self) -> None: ...

class FaceRecognizerPort(Protocol):
    def predict(self, face_gray: ArrayLike) -> RecognitionResult: ...
    def train(self, labeled_faces: Sequence[tuple[int, ArrayLike]]) -> None: ...
    def save(self) -> None: ...
    def load(self) -> None: ...

class DeviceTransportPort(Protocol):
    def send_access_command(self, command: DeviceCommand) -> None: ...
    def poll(self) -> list[DeviceEvent]: ...
    def health(self) -> TransportHealth: ...

```

```

class IdentityRepository(Protocol):
    def get_card_owner(self, uid: str) -> Person | None: ...
    def get_person(self, person_id: str) -> Person | None: ...
    def create_person(self, person: Person) -> None: ...
    def bind_card(self, uid: str, person_id: str) -> None: ...

class EventRepository(Protocol):
    def append(self, event: AccessEvent) -> None: ...

class ClockPort(Protocol):
    def now_monotonic(self) -> float: ...

```

4. Core Authorization Policy

```

def authorize(session, recognition, now, config):
    if now >= session.deadline:
        return DENY(TIMEOUT)

    if recognition is MULTIPLE_FACES:
        return DENY(MULTIPLE_FACES)

    if recognition is NONE:
        return CONTINUE

    if recognition.distance > config.lbph_threshold:
        return CONTINUE

    if recognition.person_id != session.expected_person_id:
        return DENY(FACE_MISMATCH)

    session.match_count += 1
    if session.match_count >= config.required_consistent_matches:
        return GRANT(MATCHED_OWNER)

    return CONTINUE

```

Important: RecognitionResult must be data. It must not call the servo, LEDs, buzzer, serial adapter, or database.

5. AccessController Pseudocode

```

loop:
    events = device_transport.poll()

    for event in events:
        if event.type == RFID_DETECTED:
            begin_or_reject_session(event)

```

```

if session.state == VERIFYING:
    if clock.now_monotonic() >= session.deadline:
        deny(TIMEOUT)
    else:
        frame = camera.read()
        faces = detector.detect(frame)

        if len(faces) == 0:
            continue
        if len(faces) > 1:
            deny(MULTIPLE_FACES)
        else:
            result = recognizer.predict(normalize(faces[0]))
            decision = policy.evaluate(session, result)
            execute_decision(decision)

```

6. Decision Execution

GRANT

- > persist AccessEvent(GRANTED)
- > device.send_access_command(decision=grant, hold_ms=config.door_hold_s)
- > wait for command_result
- > if execution fails: log ACTUATOR_FAILURE; keep safe state

DENY / TIMEOUT / FAULT

- > persist AccessEvent(result)
- > device.send_access_command(decision=deny, indication=reason)
- > remain closed

7. NDJSON Protocol Contract

Envelope

```

{
  "v": 1,
  "type": "...",
  "event_id": "...", // event messages
  "session_id": "...", // access workflow correlation
  "command_id": "...", // device command correlation
  "device_id": "door-01",
  "payload": {}
}

```

RFID_DETECTED payload:

```

{"uid":"A1B2C3D4"}

```

ACCESS_COMMAND payload:

```

{"decision":"grant|deny","reason":"MATCHED_OWNER|FACE_MISMATCH|TIMEOUT|...", "door_hold_ms":3000}

```

COMMAND_RESULT payload:

```
{"result": "executed | rejected | error", "error_code": null}
```

HEARTBEAT payload:

```
{"state": "CLOSED | OPEN | FAULT"}
```

8. Device Firmware State Machine

DEVICE_STARTUP

- > CLOSED_SAFE

- > IDLE

IDLE + RFID card -> emit RFID_DETECTED -> remain IDLE

IDLE + ACCESS_COMMAND(grant)

- > validate command/session freshness

- > OPENING

- > OPEN_HOLD

- > CLOSING

- > IDLE

IDLE + ACCESS_COMMAND(deny)

- > DENY_INDICATION

- > IDLE

Any invalid/stale/duplicate command

- > ignore or reject

- > CLOSED_SAFE

9. Hardware Pin Baseline

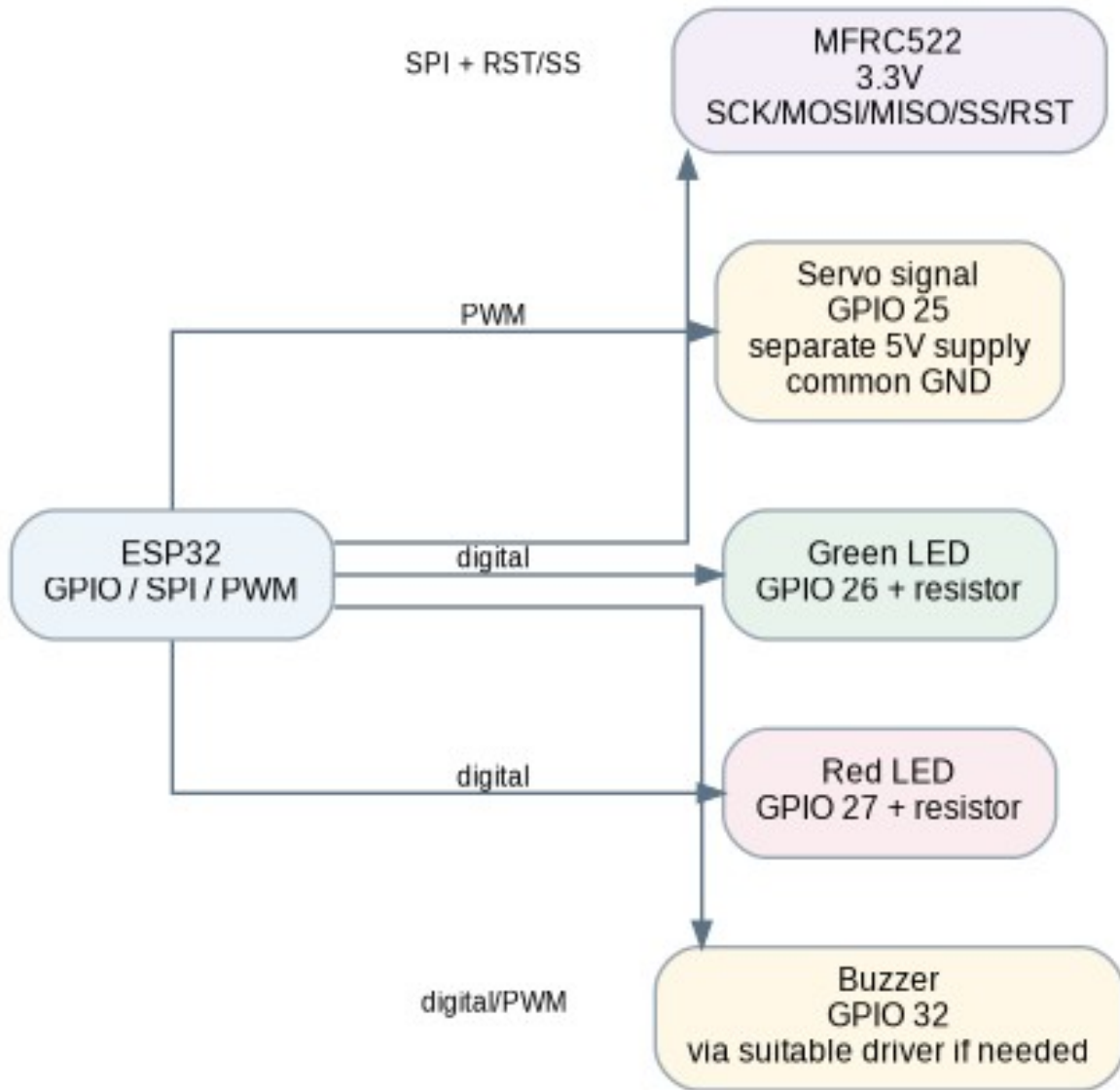


Figure 3. Proposed example pin map; verify against the exact ESP32 board before wiring

Component	Signal	ESP32 pin	Notes
MFRC522	SCK	18	SPI clock
MFRC522	MISO	19	SPI data to ESP32
MFRC522	MOSI	23	SPI data from ESP32
MFRC522	SS/SDA	5	SPI chip select
MFRC522	RST	22	Reader reset
MFRC522	3.3V/GND	3V3/GND	Keep logic at supported voltage
Servo	Signal	25	PWM signal; servo should

Component	Signal	ESP32 pin	Notes
			use suitable external power
Green LED	Anode via resistor	26	Digital output
Red LED	Anode via resistor	27	Digital output
Buzzer	Control	32	Digital or PWM depending on buzzer type; use driver if required

Board note: GPIO choices are a proposed baseline for a common ESP32 development board, not a universal wiring law. Verify pin availability/strapping conflicts on the exact board revision before assembly.

10. Face Processing Pipeline

```

capture frame
-> grayscale (for LBPH)
-> face detection
-> exactly-one-face gate
-> crop face region
-> normalize dimensions / optional histogram equalization
-> LBPH.predict()
-> threshold check
-> identity binding check
-> stable-match gate

```

OpenCV documentation states that LBPH expects grayscale image input and exposes a threshold-based rejection mechanism. The concrete detection method is an implementation choice; a lightweight local detector is sufficient for this MVP.

11. Enrollment Pipeline

```

enroll_person(person_id):
  capture until sample_count >= configured minimum
  for each frame:
    faces = detector.detect(frame)
    accept only exactly one usable face
    save normalized grayscale sample
  train LBPH on labeled samples
  persist model + label map
  bind RFID UID -> person_id
  run live verification test

```

12. Configuration

```
verification_window_s: 10

door_hold_s: 3
required_consistent_matches: 3
lbph_threshold: CALIBRATE_FROM_ACTUAL_DATA
camera_index: 0
serial_baud: 115200
heartbeat_interval_s: 1
heartbeat_timeout_s: 3
allow_multiple_faces: false
```

The LBPH threshold is not hard-coded by this document; it must be calibrated from the actual dataset and webcam, then recorded as a versioned configuration value.

13. Error / Reason Codes

Code	Meaning	Action
UNKNOWN_RFID	UID not enrolled/active	Deny
FACE_NOT_FOUND	No face in frame	Continue until timeout
MULTIPLE_FACES	Ambiguous frame	Deny
FACE_MISMATCH	Known face is not card owner	Deny
LOW_CONFIDENCE	LBPH distance above threshold	Continue until timeout
TIMEOUT	10-second window expired	Deny
CAMERA_ERROR	Webcam unavailable	Fault/deny
DEVICE_LINK_ERROR	ESP32 communication unavailable	Fault/deny
COMMAND_REJECTED	ESP32 rejected command	Fault/deny; log
ACTUATOR_FAILURE	Expected device action failed	Log; maintain safe state

14. Test Pyramid

Level	Examples	Gate
Unit	Policy decisions, state transitions, timer expiration, protocol parsing, repository rules	Every requirement in policy covered.
Integration	Camera + detector + LBPH, SQLite +	Deterministic flows work end-to-end

Level	Examples	Gate
	enrollment, serial adapter + firmware simulator	on laptop.
Hardware-in-loop	Real MFRC522, real cards, real servo/LED/buzzer, webcam	ACC-001..ACC-010 pass under controlled conditions.
Failure injection	Unplug camera, unplug serial, send malformed command, reset ESP32	Door remains closed and fault is observable.

15. Implementation Constraints for an AI Coding Agent

- Implement requirement IDs exactly as written; do not invent product features to fill perceived gaps.
- Do not introduce network services, brokers, cloud APIs, or microservices for the MVP.
- Do not let recognition code access GPIO or serial directly; use ports/adapters.
- Do not let firmware make biometric or identity decisions.
- Do not grant access on one weak signal; the canonical authorization rule must remain unchanged.
- Do not silently change the 10-second window, fail-closed rule, or wrong-face behavior.
- Every change that affects behavior must update the relevant requirement/test mapping.
- Prefer small deterministic modules with tests over framework-heavy abstractions.

16. Definition of Done

Area	Done when
Requirements	All FR/NFR/SEC requirements traced to implementation and at least one test.
Software	Host starts, initializes dependencies, runs state machine, persists identities/events, and exposes useful logs.
Firmware	ESP32 reads RFID, accepts versioned commands, drives outputs, and fails safe.
Face recognition	Enrollment and live recognition work with actual webcam and calibrated threshold.
Integration	Correct path and all negative paths in ACC-001..ACC-010 work.
Safety	Power cycle, serial loss, camera loss, malformed command, and restart leave the door closed.
Documentation	Known photo-spoof and RFID-cloning limitations are visible in project docs.

17. Reference Notes

OpenCV LBPHFaceRecognizer supports configurable radius, neighbors, grid dimensions, and threshold; its documented API also states that the LBPH features expect grayscale images. NXP currently lists the MFRC522 as End of Life / not recommended for new designs, which is acceptable here only because the part is a fixed project constraint for a prototype.